Cardiology - Catheterization and Consult Reports

Record 1

PROCEDURE PERFORMED:,1. Left heart catheterization with coronary angiography, vein graft angiography and left ventricular pressure measurement and angiography.,2. Right femoral selective angiogram.,3. Closure device the seal the femoral arteriotomy using an Angio-Seal.,INDICATIONS FOR PROCEDURE: ,The patient with known coronary atherosclerotic heart disease and multiple risk factors of coronary disease, who had her last coronary arteriogram performed in 2004. She has had complaints of progressive chest discomfort, and has ongoing risks including current smoking, diabetes, hypertension, hyperlipidemia to name a few. The decision was made to proceed on with percutaneous evaluation and possible intervention given her known disease and the possibility of disease progression.,DESCRIPTION OF PROCEDURE:, After informed consent was obtained, The patient was taken to cardiac catheterization lab where her procedure was performed. She was prepped and prepared on the table; after which, her right groin was locally anesthetized with 1% lidocaine. Then, a 6-French sheath was inserted into the right femoral artery over a standard 0.035 guide wire. Coronary angiography and left ventricular measurement and angiography were performed using a 6-French JL4 diagnostic catheter to image the left coronary artery. A 6-French JR4 diagnostic catheter to image the right groin and her artery and the saphenous vein graft conduit. Subsequently, a 6-French angled pigtail catheter was used to measure left ventricular pressures and to perform a power injection, a left ventriculogram at 8 mL per second for a total of 30 mL. At the conclusion of the diagnostic evaluation, the patient had selective arteriography of her right femoral artery, which showed the right femoral artery to be free of significant atherosclerotic plaque. Did have a normal bifurcation into the superficial femoral and profunda femoris arteries and to have an arteriotomy that was in the common femoral artery away from the bifurcation. As such, an initial attempt to advance a Perclose device failed to allow the device descend to _____ tract and into the appropriate position within the artery. As such, the Perclose was never deployed and was removed intact over the wire from the system. We then replaced this with a 6-French Angio-Seal which was used to seal the femoral arteriotomy with achievement of hemostasis. The patient was subsequently dispositioned back to the MAC Unit where she will complete her bedrest prior to her disposition to home.,HEMODYNAMIC DATA:, Opening aortic pressure 125/60, left ventricular pressure 108/4 with an end-diastolic pressure of 16. There was no significant gradient across the aortic valve on pullback from the left ventricle. Left ventricular ejection fraction was 55%. Mitral regurgitation was less than or equal to 1+. There was normal wall motion in the RAO projection.,CORONARY ANGIOGRAM:, The left main coronary artery had mild atherosclerotic plaque. The proximal LAD was 100% occluded. The left circumflex had mild diffuse atherosclerotic plaque. The obtuse marginal branch which operates as an OM-2 had a mid approximately 80% stenosis at a kink in the artery. This appears to be the area of a prior anastomosis, the saphenous vein graft to the OM. This is a very small-caliber vessel and is 1.5-mm in diameter at best. The right coronary artery is dominant. The native right coronary artery had mild proximal and mid atherosclerotic plaque. The distal right coronary artery has an approximate 40% stenosis. The posterior left ventricular branch has a proximal 50 to 60% stenosis. The proximal PDA has a 40 to 50% stenosis. The saphenous vein graft to the right PDA is widely patent. There was competitive flow noted between the native right coronary artery and the saphenous vein graft to the PDA. The runoff from the PDA is nice with the native proximal PDA and PLV disease as noted above. There is also some retrograde filling of the right coronary artery from the runoff of this graft. The saphenous vein graft to the left anterior descending is widely patent. The LAD beyond the distal anastomosis is a relatively small-caliber vessel. There is some retrograde filling that allows some filling into a more proximal diagonal branch. The saphenous vein graft to the obtuse marginal was known to be occluded from the prior study in 2004. Overall, this study does not look markedly different than the procedure performed in 2004.,CONCLUSION:, 100% proximal LAD mild left circumflex disease with an OM that is a small-caliber vessel with an 80% lesion at a kink that is no amenable to percutaneous intervention. The native right coronary artery has mild to moderate distal disease with moderate PLV and PDA disease. The saphenous vein graft to the OM is known to be 100% occluded. The saphenous vein graft to the PDA and the saphenous vein graft to the LAD are open. Normal left ventricular systolic function.,PLAN:, The plan will be for continued medical therapy and risk factor modification. Aggressive antihyperlipidemic and antihypertensive control. The patient's goal LDL will

be at or below 70 with triglyceride level at or below 150, and it is very imperative that the patient stop smoking.,After her bedrest is complete, she will be dispositioned to home, after which, she will be following up with me in the office within 1 month. We will also plan to perform a carotid duplex Doppler ultrasound to evaluate her carotid bruits.

Record 2

INDICATION: ,Chest pain.,INTERPRETATION: , Resting heart rate of 71, blood pressure 100/60. EKG normal sinus rhythm. The patient exercised on Bruce for 8 minutes on stage III. Peak heart rate was 151, which is 87% of the target heart rate, blood pressure of 132/54. Total METs was 10.1. EKG revealed nonspecific ST depression in inferior and lateral leads. The test was terminated because of fatigue. The patient did have chest pain during exercise that resolved after termination of the exercise.,IN SUMMARY:,1. Positive exercise ischemia with ST depression 0.5 mm.,2. Chest pain resolved after termination of exercise.,3. Good exercise duration, tolerance and double product.,NUCLEAR INTERPRETATION:,Resting and stress images were obtained with 10.1 mCi and 34.1 mCi of tetraphosphate injected intravenously by standard protocol. Nuclear myocardial perfusion scan demonstrates homogenous and uniform distribution with tracer uptake without any evidence of reversible or fixed defect. Gated SPECT revealed normal wall motion, ejection fraction of 68%. End-diastolic volume of 77, end-systolic volume of 24.,IN SUMMARY:,1. Normal nuclear myocardial perfusion scan.,2. Ejection fraction of 68% by gated SPECT.

Record 3

REASON FOR EXAM: , Dynamic ST-T changes with angina.,PROCEDURE:,1. Selective coronary angiography.,2. Left heart catheterization with hemodynamics.,3. LV gram with power injection.,4. Right femoral artery angiogram.,5. Closure of the right femoral artery using 6-French AngioSeal.,Procedure explained to the patient, with risks and benefits. The patient agreed and signed the consent form.,The patient received a total of 2 mg of Versed and 25 mcg of fentanyl for conscious sedation. The patient was draped and dressed in the usual sterile fashion. The right groin area infiltrated with lidocaine solution. Access to the right femoral artery was successful, okayed with one attempt with anterior wall stick. Over a J-wire, 6-French sheath was introduced using modified Seldinger technique.,Over the J-wire, a JL4 catheter was passed over the aortic arch. The wire was removed. Catheter was engaged into the left main. Multiple pictures with RAO caudal, AP cranial, LAO cranial, shallow RAO, and LAO caudal views were all obtained. Catheter disengaged and exchanged over J-wire into a JR4 catheter, the wire was removed. Catheter with counter-clock was rotating to the RCA one shot with LAO, position was obtained. The cath disengaged and exchanged over J-wire into a pigtail catheter. Pigtail catheter across the aortic valve. Hemodynamics obtained. LV gram with power injection of 36 mL of contrast was obtained.,The LV gram assessed followed by pullback hemodynamics.,The catheter exchanged out and the right femoral artery angiogram completed to the end followed by the removal of the sheath and deployment of 6-French AngioSeal with no hematoma. The patient tolerated the procedure well with no immediate postprocedure complication.,HEMODYNAMICS: ,The aortic pressure was 117/61 with a mean pressure of 83. The left ventricular pressure was 119/9 to 19 with left ventricular end-diastolic pressure of 17 to 19 mmHg. The pullback across the aortic valve reveals zero gradient.,ANATOMY: ,The left main showed minimal calcification as well as the proximal LAD. No stenosis in the left main seen, the left main bifurcates in to the LAD and left circumflex.,The LAD was a large and a long vessel that wraps around the apex showed no focal stenosis or significant atheromatous plaque and the flow was TIMI 3 flow in the LAD. The LAD gave off two early diagonal branches. The second was the largest of the two and showed minimal lumen irregularities, but no focal stenosis.,Left circumflex was a dominant system supplying three obtuse marginal branches and distally supplying the PDA. The left circumflex was large and patent, 6.0 mm in diameter. All three obtuse marginal branches appeared to be with no significant stenosis.,The obtuse marginal branch, the third OM3 showed at the origin about 30 to 40% minimal narrowing, but no significant stenosis. The PDA was wide, patent, with no focal stenosis.,The RCA was a small nondominant system with no focal stenosis and supplying the RV marginal.,LV gram showed that the LV EF is preserved with EF of 60%. No mitral regurgitation identified.,IMPRESSION:,1. Patent coronary arteries with normal left anterior descending, left circumflex, and dominant left circumflex system.,2. Nondominant right, which is free of atheromatous plaque.,3. Minimal plaque in the diagonal branch II, and the obtuse,marginal branch III, with no focal stenosis.,4. Normal left ventricular function.,5. Evaluation for noncardiac chest pain would be recommended.

Record 4

REASON FOR EXAM: , Atrial flutter/cardioversion.,PROCEDURE IN DETAIL: , The procedure was explained to the patient with risks and benefits. The patient agreed and signed the consent form. The patient received sedation prior to the cardioversion with a transesophageal echo as dictated earlier with a total of 50 mcg of fentanyl and 6 mg of Versed. The pads were applied in the anterior-posterior approach. The synchronized cardioversion with biphasic energy delivered at 150 J. First attempt was unsuccessful. Second attempt at 200 J with anterior-posterior approach. With biphasic synchronized energy delivered was also unsuccessful with degeneration of the atrial flutter into atrial fibrillation.,The patient was decided to be on wave control and amiodarone and reattempted cardioversion after anticoagulation for four to six weeks and because of the reduced LV function, the success of the rate without antiarrhythmic may be low.,IMPRESSION: , Unsuccessful direct current cardioversion with permanent atrial fibrillation.

Record 5

EXAM: , Left Heart Catheterization,REASON FOR EXAM:, Chest pain, coronary artery disease, prior bypass surgery.,INTERPRETATION: , The procedure and complications were explained to the patient in detail and formal consent was obtained. The patient was brought to the cath lab. The right groin was draped in the usual sterile manner. Using modified Seldinger technique, a 6-French arterial sheath was introduced in the right common femoral artery. A JL4 catheter was used to cannulate the left coronary arteries. A JR4 catheter was used to cannulate the right coronary artery and also bypass grafts. The same catheter was used to cannulate the vein graft and also LIMA. I tried to attempt to cannulate other graft with Williams posterior catheter and also bypass catheter was unsuccessful. A 6-French pigtail catheter was used to perform left ventriculography and pullback was done. No gradient was noted. Arterial sheath was removed. Hemostasis was obtained with manual compression. The patient tolerated the procedure very well without any complications.,FINDINGS:,1. Native coronary arteries. The left main is patent. The left anterior descending artery is not clearly visualized. The circumflex artery appears to be patent. The proximal segment gives rise to small caliber obtuse marginal vessel.,2. Right coronary artery is patent with mild distal and mid segment. No evidence of focal stenosis or dominant system.,3. Bypass graft LIMA to the left anterior descending artery patent throughout the body as well the anastomotic site. There appears to be possible _____ graft to the diagonal 1 vessel. The distal LAD wraps around the apex. No stenosis following the anastomotic site noted.,4. Vein graft to what appears to be obtuse marginal vessel was patent with a small caliber obtuse marginal 1 vessel.,5. No other bypass grafts are noted by left ventriculography and also aortic root shot.,6. Left ventriculography with an ejection fraction of 60%.,IMPRESSION:,1. Left coronary artery disease native.,2. Patent vein graft with obtuse marginal vessel and also LIMA to LAD. _____ graft to the diagonal 1 vessel.,3. Native right coronary artery is patent, mild disease.,RECOMMENDATIONS: , Medical treatment.

Record 6

ASH SPLIT VENOUS PORT,PROCEDURE DETAILS: ,The patient was taken to the operating room and placed in supine position and monitored anesthesia care provided by the anesthetist. The right anterior chest and supraclavicular fossa area, neck, and left side of chest were prepped with Betadine and draped in a sterile fashion. Xylocaine 1% was infiltrated in the supraclavicular area and anterior chest along the planned course of the catheter. The patient was placed into Trendelenburg position.,The right internal jugular vein was accessed by a supraclavicular 19-gauge, thin-walled needle as demonstrated by easy withdrawal of venous blood on the first pass of the needle. Under fluoroscopic control, a J-wire was advanced into the right atrium. The needle was removed and the skin puncture site enlarged to about 8 mm with the scalpel. A second incision was made 5 cm inferior to the right midclavicular line, through which an Ash split catheter was advanced, using the tunneling rod, in a gently curving pass to exit the skin of the neck incision. The tunneling needle was removed and the catheter split up to the marker as indicated in the recommended use of the catheter.,Sequential dilators were advanced over the J-wire under fluoroscopic control to dilate the subcutaneous tunnel followed by advancement of a dilator and sheath into the right superior vena cava under fluoroscopic control. The dilator and wire were removed, leaving the sheath in position, through which a double-lumen catheter was advanced into the central venous system. The sheath was peeled away, leaving the catheter into position. Each port of the catheter was flushed with dilute heparinized saline.,The patient was returned to the flat position. The catheter was secured to the skin of the anterior chest using 2-0 Ethilon suture placed through the suture "wings.",The neck incision was closed with 3-0 Vicryl subcuticular closure and pressure dressing applied. Fluoroscopic examination of the chest revealed no evidence of pneumothorax upon completion of the procedure and the catheter was in excellent position.,The patient was returned to the recovery room for postoperative care.

Record 7

CHIEF COMPLAINT: , Chest pain.,HISTORY OF PRESENT ILLNESS:, The patient is a 40-year-old white male who presents with a chief complaint of "chest pain".,The patient is diabetic and has a prior history of coronary artery disease. The patient presents today stating that his chest pain started yesterday evening and has been somewhat intermittent. The severity of the pain has progressively increased. He describes the pain as a sharp and heavy pain which radiates to his neck & left arm. He ranks the pain a 7 on a scale of 1-10. He admits some shortness of breath & diaphoresis. He states that he has had nausea & 3 episodes of vomiting tonight. He denies any fever or chills. He admits prior episodes of similar pain prior to his PTCA in 1995. He states the pain is somewhat worse with walking and seems to be relieved with rest. There is no change in pain with positioning. He states that he took 3 nitroglycerin tablets sublingually over the past 1 hour, which he states has partially relieved his pain. The patient ranks his present pain a 4 on a scale of 1-10. The most recent episode of pain has lasted one-hour.,The patient denies any history of recent surgery, head trauma, recent stroke, abnormal bleeding such as blood in urine or stool or nosebleed.,REVIEW OF SYSTEMS:, All other systems reviewed & are negative.,PAST MEDICAL HISTORY:, Diabetes mellitus type II, hypertension, coronary artery disease, atrial fibrillation, status post PTCA in 1995 by Dr. ABC.,SOCIAL HISTORY: , Denies alcohol or drugs. Smokes 2 packs of cigarettes per day. Works as a banker.,FAMILY HISTORY: , Positive for coronary artery disease (father & brother).,MEDICATIONS: , Aspirin 81 milligrams QDay. Humulin N. insulin 50 units in a.m. HCTZ 50 mg QDay. Nitroglycerin 1/150 sublingually PRN chest pain.,ALLERGIES: , Penicillin.,PHYSICAL EXAM: , The patient is a 40-year-old white male.,General: The patient is moderately obese but he is otherwise well developed & well nourished. He appears in moderate discomfort but there is no evidence of distress. He is alert, and oriented to person place and circumstance. There is no evidence of respiratory distress. The patient ambulates

Record 8

REASON FOR CONSULT: , A patient with non-Q-wave myocardial infarction.,HISTORY OF PRESENT ILLNESS: , The patient is a pleasant 52-year-old gentleman with a history of diabetes mellitus, hypertension, and renal failure, on dialysis, who presented with emesis, dizziness, and nausea for the last few weeks. The patient reports having worsening emesis and emesis a few times. No definite chest pains. The patient is breathing okay. The patient denies orthopnea or PND.,PAST MEDICAL HISTORY:,1. Diabetes mellitus.,2. Hypertension.,3. Renal failure, on dialysis.,MEDICATIONS:, Aspirin, Coreg, doxazosin, insulin, metoclopramide, simvastatin, and Starlix.,ALLERGIES: ,NO KNOWN DRUG ALLERGIES.,SOCIAL HISTORY: , The patient denies tobacco, alcohol or drug use.,FAMILY HISTORY: , Negative for early atherosclerotic heart disease.,REVIEW OF SYSTEMS: , General: The patient denies fever or chills. Pulmonary: The patient denies hemoptysis. Cardiovascular: Refer to HPI. GI: The patient denies hematemesis or melena. The rest of systems review is negative.,PHYSICAL EXAMINATION:,VITAL SIGNS: Pulse 71, blood pressure 120/70, and respiratory rate 18.,GENERAL: A well-nourished, well-developed male in no acute distress.,HEENT: Normocephalic, atraumatic. Pupils seem to be equal, round, and reactive. Extraocular muscles are full, but the patient has left eye ptosis.,NECK: Supple without JVD or lymphadenopathy.,LUNGS: Clear to auscultation bilaterally.,CARDIOVASCULAR: PMI is displaced 0.5 cm lateral to the midclavicular line. Regular rate and rhythm, S1, S2. No definite S3, 2/6 holosystolic murmur at the apex radiating to the axilla.,ABDOMEN: Positive bowel sounds, nondistended and nontender. No hepatosplenomegaly.,EXTREMITIES: Trace pedal edema.,EKG shows atrial fibrillation with rapid ventricular response at 164 with old anteroseptal myocardial infarction and old inferior wall myocardial infarction. Subsequent EKG in sinus rhythm shows sinus rhythm with old inferior wall myocardial infarction and probable anteroseptal myocardial infarction with Q-waves in V1, V2, and up to V3.,LABORATORY EXAM: , WBC 28,800, hemoglobin 13.6, hematocrit 40, and platelets 266,000. PT 11.3, INR 1.1, and PTT 24.1. Sodium 126, potassium 4.3, chloride 86, CO2 26, glucose 371, BUN 80, and creatinine 8.4. CK was 261, then 315, and then 529 with CK-MB of 8.06, then 8.69, and then 24.6. Troponin was 0.051, then 0.46, and then 19.8 this morning.,IMPRESSION:,1. Paroxysmal atrial fibrillation. The heart rate was slowed down with IV Cardizem, the patient converted to sinus rhythm. The patient is currently in sinus rhythm.,2. Emesis. The etiology is unclear. The patient reports that the emesis is better. The patient is just having some nausea.,3. Non-Q-wave myocardial infarction. EKG shows atrial fibrillation with old anteroseptal myocardial infarction and old inferior wall myocardial infarction.,4. Diabetes mellitus.,5. Renal failure.,6. Hypertension.,7. Hypercholesterolemia.,PLAN:,1. We will start amiodarone to keep from going back into atrial fibrillation.,2. Echocardiogram.,3. Aspirin and IV heparin.,4. Serial CK-MB and troponin.,5. Cardiac catheterization, possible percutaneous coronary intervention. The risks, benefits, and alternatives were explained to the patient through a translator. The patient understands and wishes to proceed.,6. IV Integrilin.